Bit.

Manipulation

Decimal to Binary

2	65	
2	32	1
2	16	0
2	8	0
2	4	0
2	2	0
	1	0

$(65)_{10} = (1000001)_2$

2	18	
2	9	0
2	4	1
2	2	0
	1	0

$(18)_{10} = (10010)_2$

Binary to Decimal

4 3 2 1 0

1 0 1 1 0 $\longrightarrow$ $1\times2^4 + 0\times2^3 + 1\times2^2 + 1\times2^1 + 0\times2^0$

$= 16 + 4 + 2 = 22$

i

1	0	1	1	0

K=1

sum += K*arr[i]

K*=2

Binary to Decimal

64	32	16	8	4	2	1
1	0	0	0	1	0	1

$$= 64 \cdot 1 + 4 \cdot 1 + 1 \cdot 1 = 69$$

Binary to Decimal

000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15

$13 = 8 + 4 + 1$

Bitwise &, Bitwise |

and

or

$1 \& 1 = 1$

$1 \& 0 = 0$

$0 \& 1 = 0$

$0 \& 0 = 0$

$1 | 1 = 1$

$1 | 0 = 1$

$0 | 1 = 1$

$0 | 0 = 0$

Bitwise &, Bitwise |

$$
\begin{array}{r}
1\ 1\ 0\ 0\ 1\ 1 \\
\& \quad 1\ 0\ 1\ 0\ 1\ 0 \\
\hline
1\ 0\ 0\ 0\ 1\ 0
\end{array}
\qquad
\begin{array}{r}
51 \\
\& \ 42 \\
\hline
34
\end{array}
$$

$$
\begin{array}{r}
1\ 1\ 0\ 0\ 1\ 1 \\
| \quad 1\ 0\ 1\ 0\ 1\ 0 \\
\hline
1\ 1\ 1\ 0\ 1\ 1
\end{array}
\qquad
\begin{array}{r}
51 \\
| \ 42 \\
\hline
59
\end{array}
$$

$1 \mid x = 1$

$0 \,\&\, x = 0$

$x \in \{0, 1\}$

0 false off

1 true on set

Bitwise $\wedge$ (XOR)

$$a \,\&\, b = b \,\&\, a$$
$$a \mid b = b \mid a$$
$$a \wedge b = b \wedge a$$

$$1 \wedge 1 = 0$$
$$1 \wedge 0 = 1$$
$$0 \wedge 1 = 1$$
$$0 \wedge 0 = 0$$

$$0 \wedge x = x$$

$$x \in \{0, 1\}$$

Ex: $\underbrace{\underbrace{1 \wedge 1}_{0} \wedge 0}_{0}$

$$\begin{array}{r} 1\ 0\ 0\ 1\ 1\ 1 \\ \wedge\ 1\ 0\ 0\ 1\ 1\ 1 \\ \hline 0\ 0\ 0\ 0\ 0\ 0 \\ \hline \end{array}$$

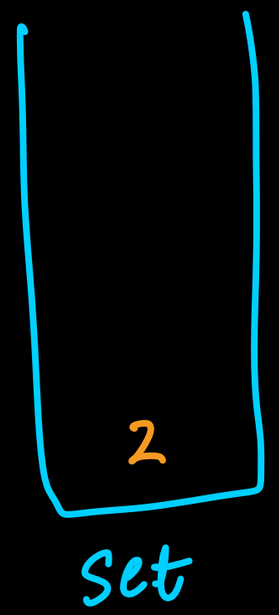
$a \wedge a = 0$ where $a \in \mathbb{Z}$

Ques: Single Number

arr = { 1, 2, 3, 1, 4, 4, 3 }

Hint: $a \wedge a = 0$

$0 \wedge b = b$



T.C. = O(n)

A.S. = O(n)

$1 \wedge 2 \wedge 3 \wedge 1 \wedge 4 \wedge 4 \wedge 3 = 2$

$2 \wedge 0 \wedge 0 \wedge 0 = 2$

Ques: Swap 2 Numbers

$a = 3, \; b = 8$

$a = a \wedge b$ $\qquad a = 3 \wedge 8 \quad b = 8$

$b = a \wedge b$ $\qquad a = 3 \wedge 8 \quad b = (3 \wedge 8) \wedge 8 \rightarrow b = 3$

$a = a \wedge b$ $\qquad a = 3 \wedge 8 \wedge 3 \quad b = 3$

1's Complement (~), 2's Complement

flip all digits

$\sim x + 1$

$-x$

or

1 0 1 1 0 1

0 1 0 0 1 0

$\sim x$

4 bits

32 bits

nibble, byte, short, int, long

how -ve numbers are stored

$$\frac{2^{32}}{2} = \cancel{2^{16}}\ 2^{31}$$

1 byte = 8 bits

$$\underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} = 2^8 \text{ numbers } (256)$$

byte range → -128 to 127 or -2^7 to $2^7 - 1$

int range → -2^{31} to $2^{31} - 1$

nibble, byte, short, int, long

how -ve numbers are stored

byte x = 23

0 0 0 1 0 1 1 1

$\downarrow \sim x$

byte y = -23

```
  1 1 1 0 1 0 0 0
+ 0 0 0 0 0 0 0 1
  ---------------
  1 1 1 0 1 0 0 1
```

byte range

-128 to 127

$-x = 2^s$ complement of x

$-x = \sim x + 1$

Binary to Decimal (-ve)

$\boxed{1\ 0\ 1\ 1\ 0\ 1\ 1\ 0}$ $= -x$ or $\sim x + 1$ → -74

$1\ 0\ 1\ 1\ 0\ 1\ 0\ 1 = \sim x$

$0\ 1\ 0\ 0\ 1\ 0\ 1\ 0 = x$

$\underset{\text{MSB}}{\underline{1}}\ 0\ 1\ 1\ 0\ 1\ \underset{\text{LSB}}{\underline{1}}$

Behavior of operators

$$\left.\begin{aligned} 1 \mid x &= 1 \\ 0 \mid x &= x \\ 0 \,\&\, x &= 0 \\ 1 \,\&\, x &= x \\ 0 \wedge x &= x \end{aligned}\right] \quad x \in \{0, 1\}$$

$$\left.\begin{aligned} a \wedge a &= 0 \\ a \wedge 0 &= a \end{aligned}\right] \quad a \in \mathbb{Z}$$

$$-x = \sim x + 1$$

<< and >> operators

left shift

right shift

$x = 1\ 0\ 1\ 1\ 0$

$x >> 1 = 1\ 0\ 1\ 1$

$x >> 3 = 1\ 0$

$x << 1 = 1\ 0\ 1\ 1\ 0\ 0$

$x << 2 = 1\ 0\ 1\ 1\ 0\ 0\ 0$

Ques: 2's Power

Given (n). Find 2^n in O(1) time

$2^0 = 1$

$2^1 = 10$

$2^2 = 100$

$2^3 = 1000$

$2^4 = 10000$

$2^n = (1 << n)$

Ques: Check if Kth bit is set or not

500 → 1 1 1 1 1 0 1 0 0 $\quad$ K = 3

3 2 1 0

n >> K , 500 >> 3

1 1 1 1 1 0

```
return ((n>>K)%2 == 1);
```

Ques: Turn on / Set the Kth bit

$$
\begin{array}{r}
1\,1\,1\,1\,1\,\underline{0}\,1\,0\,0 \\
|\quad 0\,0\,0\,0\,0\,1\,0\,0\,0 \\
\hline
1\,1\,1\,1\,1\,\underline{1}\,1\,0\,0 \\
\hline
\end{array}
$$

$K = 3$

$1 << K$

Concept of creating a bit-mask

```
return (n | (1<<K));
```

Ques: Turn off / Clear the Kth bit

$$
\begin{array}{rl}
 & 1\ 0\ 1\ \underline{1}\ 0\ 0\ 1\ 1\ 0\ 1 \\
\& & 1\ 1\ 1\ 0\ 1\ 1\ 1\ 1\ 1\ 1 \\
\hline
 & 1\ 1\ 1\ \underline{0}\ 0\ 0\ 1\ 1\ 0\ 1 \\
\hline
\end{array}
$$

$K = 6$

$\longrightarrow \sim(1<<K)$

On $\rightarrow$ |

Off $\rightarrow$ &

Toggle $\rightarrow$ ^

```
return n & (~(1<<K));
```

Ques: Toggle the Kth bit

$$
\begin{array}{r}
1\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ 1 \\
\wedge\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0 \\
\hline
1\ 0\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1
\end{array}
$$

$K = 5$

$(1 << K)$

$0 \wedge x = x$

```
return n ^ (1<<K);
```

Ques: Rightmost set bit

Ques: Power of Two

Ques: Number of 1 Bits

Ques: Minimum bit flips to convert number

Ques: Subsets

Ques: Power of Four

Ques: Score after Flipping Matrix

Ques: Single Number II

Ques: Single Number III

Ques: Reverse Bits

Ques: XOR of numbers in a given range

Ques: XOR queries of a Subarray

Ques: XOR operation in an Array

THANKYOU

Cuties